wisdomik/QuiltNet-B-32

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Zero-Shot Image Classification | OpenCLIP | PyTorch | clip | vision | language | histopathology | histology | medical | arxiv:2306.11207

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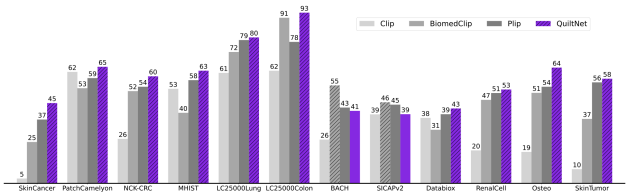
Model card | Files and versions xet | Community 2 | Use this model

QuiltNet-B-32 Description

QuiltNet-B-32 is a CLIP ViT-B/32 vision-language foundation model trained on the Quilt-1M dataset curated from representative histopathology videos. It can perform various vision-language processing (VLP) tasks such as cross-modal retrieval, image classification, and visual question answering. QuiltNet establishes new state of the art in a wide range of standard datasets, and substantially outperforms prior VLP approaches:



Citation

```
@misc{ikezogwo2023quilt1m,
      title={Quilt-1M: One Million Image-Text Pairs
      author={Wisdom Oluchi Ikezogwo and Mehmet Sayg
      year={2023},
      eprint={2306.11207},
      archivePrefix={arXiv},
      primaryClass={cs.CV}
}
```

Uses

As per the original OpenAI CLIP model card, this model is intended as a research output for research communities. We hope that this model will enable researchers to better understand and explore zero-shot, arbitrary image classification. We also hope it can be used for interdisciplinary studies of the potential impact of such model.

The OpenAI CLIP paper includes a discussion of potential downstream impacts to provide an example for this sort of

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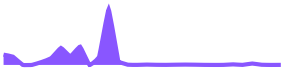

Inference Providers NEW

Zero-Shot Image Classification

This model isn't deployed by any Inference Provider.

Ask for provider support

Spaces using `wisdomik/QuiltNet-B-32` 2

Dggatsby/wisdomik-QuiltNet-B-32

saradha12/wisdomik-QuiltNet-B-32

Paper for `wisdomik/QuiltNet-B-32`

Quilt-1M: One Million Image-Text Pairs for Histopathology

Paper • 2306.11207 • Published Jun 20, 2023 • 1

analysis.

Direct Use

Zero-shot image classification, image and text retrieval, among others.

Downstream Use

Image classification and other image task fine-tuning, linear probe image classification, image generation guiding and conditioning, among others.

Intended Use

The model is intended as a research output for research communities. We hope that this model will enable researchers to better understand and explore zero-shot, arbitrary image classification. We also hope it can be used for interdisciplinary studies of the potential impact of such models.

Primary intended uses

The primary intended users of these models are AI researchers.

We primarily imagine the model will be used by researchers to better understand robustness, generalization, and other capabilities, biases, and constraints of computer vision histopathology models.

Out-of-Scope Use Cases

Any deployed use case of the model - whether commercial or not - is currently out of scope. Non-deployed use cases such as image search in a constrained environment, are also not recommended unless there is thorough in-domain testing of the model with a specific, fixed class taxonomy.

Since the model has not been purposefully trained in or evaluated on any languages other than English, its use should be limited to English language use cases.

Further the above notice, the Quilt-1M dataset used in training of these models has additional considerations, see below.

Training Data

This model was trained with QUILT-1M is an image-text dataset for histopathology. Curated from educational videos on Youtube QUILT-1M contributes the largest dataset for vision language modeling in histopathology.

IMPORTANT NOTE: The motivation behind dataset creation is to democratize research and experimentation around large-scale multi-modal model training and handling of uncurated, large-scale histopathology datasets crawled from publically available internet. Our recommendation is therefore to use the dataset for research purposes.

Evaluation

Evaluation done with code in the CLIP Benchmark suite and results can be found in the paper on a list of varying histology tasks and datasets.

Disclaimer

It is important to note that the results obtained from this function are not intended to constitute medical advice or replace consultation with a qualified medical professional. The use of this function is solely at your own risk and should be consistent with applicable laws, regulations, and ethical considerations. We do not warrant or guarantee the accuracy, completeness, suitability, or usefulness of this function for any particular purpose, and we hereby disclaim any liability arising from any reliance placed on this function or any results obtained from its use.

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